

Short-Term $PM_{2.5}$ Forecasting in Yerevan: Comparing Statistical and Machine Learning Models

Anonymous author(s)

Author identities withheld for double-blind review (ICDSG 2026)

Abstract—We compare parsimonious time-series models (AR, SARIMA) to machine learning (gradient boosting, DeepAR) for hourly $PM_{2.5}$ in Yerevan, Armenia, at horizons 1–24 h, with strict chronological splits, walk-forward robustness, block-bootstrap confidence intervals for MAE, and Diebold–Mariano tests with HAC variance. While AR(24) provides the best performance for spatially aggregated city and district metrics—including district-weighted MAE $3.66 \mu\text{g}/\text{m}^3$ versus persistence at $3.82 \mu\text{g}/\text{m}^3$ ($h=1$)—simple Persistence remains the most robust baseline for individual station-level forecasting. Tuned ensembles improve over ridge but do not beat strong linear autoregression at 1 h on the primary test window. DeepAR (MAE $26.31 \mu\text{g}/\text{m}^3$ at 1 h) under a deliberately minimal, “vanilla” configuration performs poorly, illustrating that complex sequence models are not informative without task-specific scaling and tuning in high-variance urban series. Implementation details are omitted for double-blind review and will be documented in camera-ready materials.

Index Terms—Air quality, $PM_{2.5}$, forecasting, Diebold–Mariano test, machine learning, time series.

I. INTRODUCTION

Fine particulate matter ($PM_{2.5}$) is a major urban health risk and a regulated indicator in many jurisdictions [1]. Short-horizon forecasts support public alerts, routing advice, and communication when concentrations rise quickly. Yerevan, the capital of Armenia, exhibits strong winter pollution episodes tied to space heating, topography, and meteorology; at the same time, open national monitoring supports reproducible modeling from public feeds.

Machine learning is widely applied to air quality prediction [2], yet low-order autoregressive and seasonal ARIMA structures often remain strong baselines when persistence is high and samples are limited [3], [4]. Fair comparison therefore requires leakage control (no future covariates in features), temporal robustness beyond a single split, and uncertainty for point metrics, not only leaderboard MAE.

Validation hygiene matters: leakage and single-fold leaderboards inflate ML gains [4], and spatial smoothing changes error scales, so scores must name their geographic support. We therefore prioritize chronological splits and walk-forward re-fitting to reduce “lucky split” bias that is common in air-quality ML comparisons.

We ask: (i) when linear models outperform ML at 1–24 h; (ii) how city, district, and station aggregation changes rankings; (iii) whether tuned boosting and a sequence model (DeepAR) change conclusions; (iv) whether rankings hold

under walk-forward refitting. We report frozen AR coefficients after training, HAC Diebold–Mariano tests [5], [6], circular block bootstrap for MAE uncertainty, and a modular, scripted pipeline for replication.

Contributions. (1) Preprocessing with controlled imputation and strictly past-only features. (2) A benchmark zoo: baselines, AR(24), SARIMA, ridge, forests, boosted trees (Hyperopt-tuned), inverse-MAE and average ensembles, and DeepAR. (3) Walk-forward evaluation with SARIMA re-estimated each fold. (4) Spatial comparison of weighted test MAE across aggregation levels.

II. RELATED WORK

Forecast comparison tests [5], [6] motivate paired testing on loss differentials rather than informal gap inspection: when errors are serially correlated, standard normal critical values for mean error gaps are misleading, whereas HAC variance estimates align nominal size better in hourly environmental series. Environmental ML surveys [2] document widespread nonlinear modeling while often leaving classical time-series baselines under-tuned or mis-aligned with the forecast horizon; a common pattern is to compare a deep model to a default ARIMA rather than to a long-lag AR that matches the effective memory of the target.

Direct multi-step estimation [7] matches our per-horizon supervised learners. Local context appears in [8]; aggregating stations to districts or a city mean changes noise and can reorder models, which matters for mapped indicators versus sensor-level analytics.

III. DATA AND METHODS

A. Data, stations, and spatial targets

Open hourly air quality from the national portal is aligned to a regular clock; the primary target is city-average $PM_{2.5}$ ($\mu\text{g}/\text{m}^3$). Covariates include PM_{10} , temperature, pressure, and humidity when complete on the modeling grid. The district inventory lists 205 unique monitoring stations that enter the city mean. Figure 1 plots every geocoded site used in the analysis; district labels report site counts that *sum to 205*, matching the inventory total. Overlapping markers at identical coordinates can make the network appear visually sparser than the true count. Stations map to administrative districts (eleven of twelve districts represented; one district has no assigned sites). District-level series are district means over

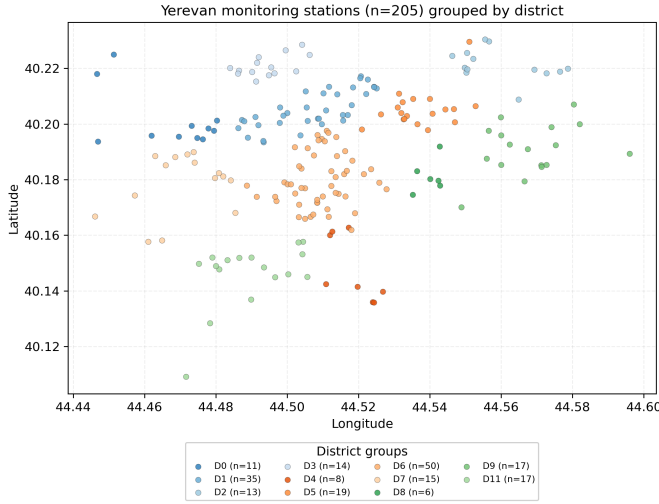


Fig. 1. Geocoded monitoring stations ($n=205$; district legend n sum to 205) used in this study, colored by district (metadata coordinates).

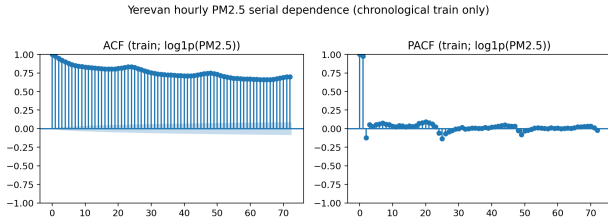


Fig. 2. ACF and PACF of training-window city-mean $PM_{2.5}$ (hourly).

member stations; station-level analyses weight units by inverse historical missingness so that informative sites contribute more to aggregates of errors.

Figure 1 situates the monitoring network geographically; Section IV reports accuracy on the city mean and on district- and station-weighted targets.

B. Preprocessing, gaps, and features

Short gaps in $PM_{2.5}$ use time-local interpolation; medium gaps use hour-of-day medians learned on training data; residual covariate holes use capped fills. The modeling table is restricted to rows with listwise-complete features after imputation. Feature construction uses only information available at forecast time t : lags up to 72 h, rolling means and standard deviations over 6–48 h, low-order Fourier seasonality, calendar and heating-season indicators, and optional meteorology interactions. $PM_{2.5}$ inputs are winsorized at the training 99.5th percentile to limit leverage from spikes.

Figure 2 shows autocorrelation structure on the training window for the city mean: strong short-lag dependence motivates AR(24) as a substantive baseline rather than a strawman, and periodic spikes at 24 h multiples reflect the diurnal cycle that seasonal naïve rules target.

C. Model families

Baselines. Persistence ($\hat{y}_{t+h} = y_t$), seasonal naïve with 24 h lag, and a 24 h trailing mean.

AR(24). Univariate autoregression with fixed order 24 on the target scale being modeled:

$$y_t = \sum_{i=1}^{24} \phi_i y_{t-i} + \varepsilon_t. \quad (1)$$

After joint least squares on the training window, coefficients $\{\phi_i\}$ are frozen and applied on validation and test (no recursive re-fit each hour).

SARIMA. Seasonal ARIMA $(2, 1, 2) \times (1, 0, 0)_{24}$ estimated on a trailing 8,760 h ($\approx$ one year) window before each forecast origin for scalability and recency; multi-step point forecasts use the fitted state for $h=1$ in the main table, with walk-forward re-estimation each fold.

Supervised ML. Ridge regression on the engineered lag library; random forest, XGBoost, and LightGBM with validation-tuned hyperparameters (10 Hyperopt trials in the reported run). Ensembles combine the top three validation models by inverse MAE or simple average.

DeepAR. We include a *vanilla* DeepAR baseline (Neural-Forecast) with library-default-style training and scaling, intended not as a best-effort neural competitor but to demonstrate the risk of deploying complex sequence models without task-specific preprocessing, hyperparameter search, and stability checks. The resulting 1 h MAE (Table I) is far above classical baselines and should be read as a cautionary negative result rather than a fair ceiling for deep forecasting.

D. Splits, walk-forward, and testing

The primary experiment uses a 70%/15%/15% chronological split on the modeling table: training through 2023-12-29, validation through 2025-01-14, and test through 2026-01-31 (illustrative dates from the frozen export). An expanding walk-forward scheme advances by roughly one month, refitting all models each fold; tree methods reuse validation-tuned hyperparameters for computational stability. This re-fit protocol is a deliberate guard against optimistic one-window results: if ranking changes sharply across folds, we treat single-split wins as unstable.

E. Metrics and inference

Point metrics are MAE, RMSE, and R^2 on the test slice. Supervised models are trained for horizons $\mathcal{H} = \{1, 2, 3, 4, 6, 12, 24\}$ (direct multi-step). Uncertainty for MAE uses a circular block bootstrap (2000 draws) with block length chosen to respect diurnal structure.

For paired comparison of absolute errors $\{|e_{t,h}^{(A)}|, |e_{t,h}^{(B)}|\}$, Diebold–Mariano uses mean loss differential $\bar{d}$ with a heteroskedasticity- and autocorrelation-consistent (HAC) variance estimator; we report the standard normal statistic and two-sided p -values [5], [6]. Only a small set of pairs is tested to limit multiplicity.

IV. RESULTS

A. City-mean accuracy at 1 h

Table I ranks models on the primary city-average target (all MAE values in $\mu\text{g}/\text{m}^3$). At $h=1$, AR(24) achieves MAE 2.41, Persistence 2.75, SARIMA 4.38, and Ridge 4.66—values mirrored in Table III (Table III) and the city row of Table IV (Table IV). AR(24) leads overall, followed by persistence and inverse-MAE ensembles; gradient boosting improves markedly with tuning but does not close the gap to AR(24). SARIMA and Ridge sit mid-pack; seasonal naive and DeepAR illustrate failure modes (misaligned seasonality and a minimal sequence baseline, respectively). The poor performance of DeepAR (MAE $26.31 \mu\text{g}/\text{m}^3$) highlights the sensitivity of global sequence models to the high-variance, localized spikes characteristic of Yerevan’s winter $PM_{2.5}$ pollution.

B. Bootstrap intervals and Diebold–Mariano tests

Table II pairs block-bootstrap MAE intervals with DM tests for representative comparisons at $h=1$. Intervals for AR(24) and persistence do not overlap those for ridge. On $n=9,178$ hourly test rows (HAC lag 25), the two-sided Diebold–Mariano p -values are $p<0.001$ for AR(24) versus persistence (DM=8.74) and $p<0.001$ for AR(24) versus ridge (DM=17.38); thus the city-mean MAE advantage for AR(24) is statistically discernible at conventional levels, not merely a numerical tie under serial dependence.

C. Horizons beyond one step

Figure 3 (Figure 3 in the compiled layout) traces MAE across horizons: persistence error grows faster than supervised models, while AR(24) remains strongest at short lags. Table III (Table III) gives a compact numeric view for selected models; Ridge reaches MAE $9.52 \mu\text{g}/\text{m}^3$ at 24 h, and SARIMA is reported at 1 h only ($4.38 \mu\text{g}/\text{m}^3$) because our production path emphasizes single-step state forecasts for that specification.

D. Spatial aggregation and walk-forward stability

Spatial aggregation tells a coherent story at $h=1$ (MAE in $\mu\text{g}/\text{m}^3$). On the city mean, AR(24) is best at $2.41 \mu\text{g}/\text{m}^3$ (Table IV, Table IV). On the district-weighted target (Table IV: district means with inverse-missingness weights on units), AR(24) remains best at $3.66 \mu\text{g}/\text{m}^3$, ahead of persistence at $3.82 \mu\text{g}/\text{m}^3$ (i.e., still below ≈ 4.0 but materially above the city-scale errors). At station-sample weighting, persistence wins at $9.91 \mu\text{g}/\text{m}^3$ versus AR(24) at $10.10 \mu\text{g}/\text{m}^3$ (Table IV), so finer spatial support reverses the ranking even though AR(24) still wins on smoothed city and district aggregates. Figure 4 and Figure 4 summarize 1-hour ranking and fold stability: mean ranks remain consistent with the single split—AR(24) leads among compared models on aggregated targets; SARIMA ranks between tuned gradient boosting and ridge on 1 h city MAE (Table I, Figure 4) but exhibits higher fold-to-fold variance than AR(24) in walk-forward evaluation.

V. DISCUSSION AND CONCLUSION

Strong short-lag persistence and diurnal structure make linear autoregression highly competitive on temporally honest splits; flexible ML can reduce error versus ridge yet, under our protocol, does not surpass a well-specified AR(24) on city or district-weighted 1 h MAE (district-weighted $3.66 \mu\text{g}/\text{m}^3$ for AR(24) versus $3.82 \mu\text{g}/\text{m}^3$ for persistence). While AR(24) provides the best performance for spatially aggregated city and district metrics, simple Persistence remains the most robust baseline for individual station-level forecasting (Table IV, Table IV). DeepAR in our vanilla configuration remains a deliberate negative result (Table I; MAE $26.31 \mu\text{g}/\text{m}^3$) and should not be read as a pipeline error: the failure of vanilla DeepAR is attributed to the non-stationary nature of Yerevan’s winter pollution episodes, which require custom localized scaling that global sequence models lack out-of-the-box.

Implications and future work. For city- or district-scale indicators, long-lag AR with frozen coefficients is a strong default; fine-scale analytics may need missingness-aware weighting or spatial pooling. Reporting should always state whether metrics are computed on a spatially aggregated target or on heterogeneous stations, because the “best” model can change with the support even when physics does not. Probabilistic forecasts, graph models without future leakage, and nested cross-validation are natural extensions.

Limitations. One primary hold-out plus walk-forward analysis is not full nested cross-validation; imputation injects structured assumptions; district coverage is uneven; we estimate associations for forecasting, not causal effects. Outputs are not personal exposure estimates.

Reproducibility. Data are from the national open portal (airquality.am). Software configuration, split metadata, and extended tables are withheld for double-blind review and will be included in camera-ready and/or supplemental material.

REFERENCES

- [1] World Health Organization, “Who global air quality guidelines: Particulate matter ($PM_{2.5}$ and PM_{10}), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide,” WHO guideline document, Geneva, 2021, summary guideline values for $PM_{2.5}$ exposure.
- [2] X. Shao, X. Li, X. Ma, and Y. Shi, “Predictions of $PM_{2.5}$ concentrations with machine learning techniques: A review,” *Ecological Indicators*, vol. 119, p. 106899, 2020.
- [3] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. John Wiley & Sons, 2015.
- [4] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: <https://otexts.com/fpp3/>
- [5] F. X. Diebold and R. S. Mariano, “Comparing predictive accuracy,” *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
- [6] D. I. Harvey, S. J. Leybourne, and P. Newbold, “Testing the equality of prediction mean squared errors,” *International Journal of Forecasting*, vol. 13, no. 2, pp. 281–291, 1997.
- [7] M. Marcellino, J. H. Stock, and M. W. Watson, “A comparison of direct and iterated multistep AR methods for forecasting macroeconomic time series,” *Journal of Econometrics*, vol. 135, no. 1–2, pp. 499–526, 2006.
- [8] Anonymous, “Air pollution: sources, effects, and urban monitoring context,” Technical report, 2025, affiliation withheld for double-blind review.

TABLE I
TEST PERFORMANCE AT HORIZON $h = 1$ H, CITY-AVERAGE $PM_{2.5}$ ($\mu\text{G}/\text{m}^3$).

Model	MAE	RMSE	R^2
AR(24)_frozen	2.41	4.39	0.9815
Persistence	2.75	4.93	0.9767
Ensemble_invMAE_top3	3.04	5.57	0.9702
Ensemble_avg_top3	3.23	5.94	0.9661
RandomForest_tuned	3.94	7.11	0.9515
LightGBM	4.02	7.25	0.9495
XGBoost_tuned	4.03	7.56	0.9452
LightGBM_tuned	4.05	7.45	0.9468
RandomForest	4.07	7.32	0.9486
XGBoost	4.23	8.10	0.9371
SARIMA_auto	4.38	7.67	0.9435
Ridge	4.66	8.21	0.9353
Seasonal_naive_24h	9.25	16.93	0.7250
DeepAR	26.31	41.65	-0.6639

TABLE II
LEFT: BLOCK-BOOTSTRAP 95% MAE INTERVALS. RIGHT: DIEBOLD–MARIANO TESTS ON ABSOLUTE LOSS (ILLUSTRATIVE PAIRS, $h=1$).

(a) Bootstrap MAE 95% CIs			(b) Diebold–Mariano				
Model	MAE	95% CI	Comparison	DM	p	n	HAC lag
AR(24)_frozen	2.41	[2.18, 2.66]	AR vs. persist.	8.74	<0.001	9178	25
Persistence	2.75	[2.48, 3.03]	AR vs. ridge	17.38	<0.001	9178	25
LightGBM_tuned	4.05	[3.61, 4.51]					
SARIMA_auto	4.38	[3.94, 4.82]					
Ridge	4.66	[4.19, 5.14]					

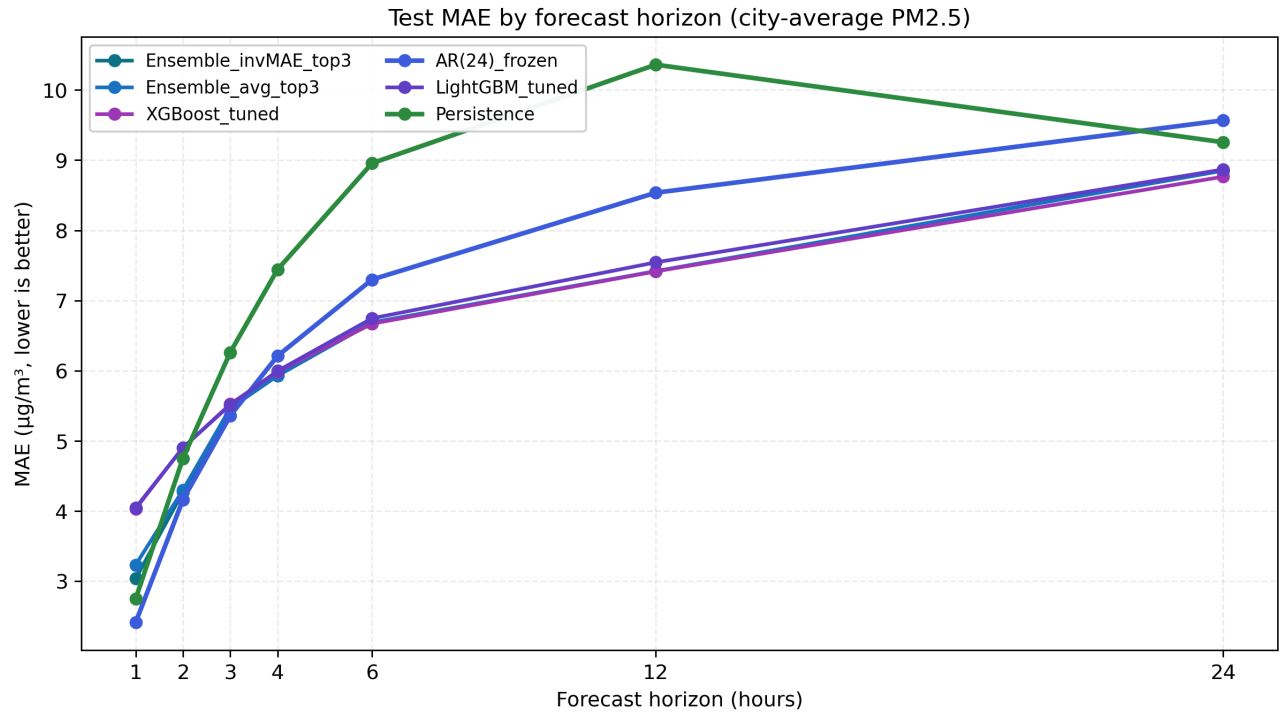


Fig. 3. Test MAE by forecast horizon for city-mean $PM_{2.5}$ (direct multi-step supervised learners; AR/SARIMA paths per implementation).

TABLE III
TEST MAE ($\mu\text{g}/\text{m}^3$) BY HORIZON FOR SELECTED MODELS. DASHES: HORIZON NOT REPORTED FOR SARIMA IN THIS CONFIGURATION.

Model	$h=1$	$h=2$	$h=4$	$h=12$	$h=24$
AR(24)_frozen	2.41	4.16	6.21	8.54	9.57
Persistence	2.75	4.75	7.44	10.36	9.26
Ensemble_avg_top3	3.23	4.30	5.93	7.42	8.85
LightGBM_tuned	4.05	4.90	6.00	7.54	8.87
SARIMA_auto	4.38	—	—	—	—
Ridge	4.66	6.10	8.07	10.01	9.52

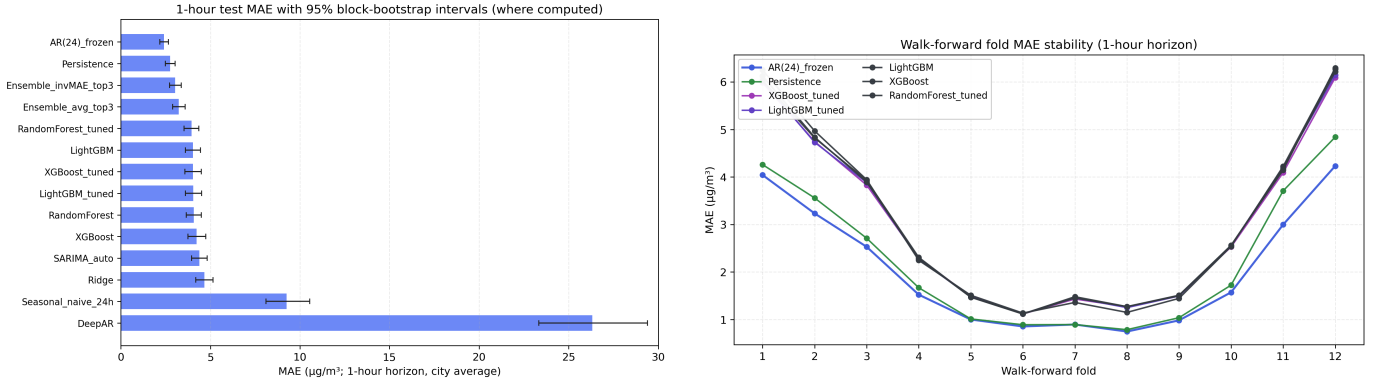


Fig. 4. Left: 1-hour test MAE (horizontal extent $\propto$ MAE, rows sorted like Table I; MAE and intervals in $\mu\text{g}/\text{m}^3$). For the top three models, bootstrap error bars overlap materially, indicating high variance in relative performance across episodes and weather regimes despite a clear ordering of point estimates. SARIMA at 4.38 lies above Ridge (4.66) in the vertical ordering (better MAE) and shows a *shorter* bar than Ridge; Seasonal Naive (9.25) is much farther out. Right: walk-forward fold MAE stability (city mean, $\mu\text{g}/\text{m}^3$).

TABLE IV
TOP-TWO MODELS BY SPATIAL AGGREGATION LEVEL ($h=1$, $\mu\text{G}/\text{M}^3$;
DISTRICT AND STATION TARGETS ARE MISSINGNESS-WEIGHTED).

Level	Model	MAE ($\mu\text{g}/\text{m}^3$)
City	AR(24)	2.41
	Persistence	2.75
District (wtd.)	AR(24)	3.66
	Persistence	3.82
Station (wtd.)	Persistence	9.91
	AR(24)	10.10